

0.3 - STATISTICS REVISION

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INTRODUCTION

In the previous chapter we covered the core machine learning topics that are key to the understanding of deep learning, but ML alone isn't a strong enough basis.

Since this field is deeply (pun intended) connected with *data analysis*, having familiarity with statistics concepts is not only recommended, but absolutely essential.

Therefore, this chapter aims at *revising known concepts* in probability and statistics, and introducing new ones that will then be used throughout the rest of the course.

CLT REVISION

It's probably worth it to start this revision from the **Central Limit Theorem (CLT)**, a crucial pillar of statistics and machine learning.

The theorem is a key concept in probability theory because it implies that probabilistic

and **statistical methods that work for normal distributions can be applicable to many problems involving other types of distributions.**

The statement is as follows:

CENTRAL LIMIT THEOREM

Let $(X_i)_{i \geq 1}$ be a sequence of **independent and identically distributed** random variables with mean μ and variance σ^2 ; and let $\bar{X}_n = \frac{1}{n}[X_1 + \dots + X_n]$ be the **sample mean** of a sample of n variables of the sequence.

Then, as n approaches infinity, the following is true:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow[n \rightarrow \infty]{} \mathcal{N}(0, 1)$$

IMPORTANT NOTE

The usefulness of the theorem stems from the fact that it **holds even if the original variables X_i themselves are not normally distributed**, greatly simplifying the analysis of random variables *for large sample sizes*.

Another crucial theorem in statistics is the (*strong*) **law of large numbers**:

(STRONG) LAW OF LARGE NUMBERS

$$P\left(\lim_{n \rightarrow \infty} \bar{X}_n = \mu\right) = 1$$

The key takeaway for us is that if we take a *sufficiently large sample size* from a *population with finite variance*, the extracted **mean of all samples $\bar{X}_n$** will be a **good approximation of the original population mean**.

HYPOTHESIS TESTING AND P VALUE

Hypothesis testing is a **formal procedure** that involves the use of statistics to investigate our ideas about the world.

Scientists most commonly use it to **test specific predictions** derived from theories, which we call hypotheses.

A key concept in hypothesis testing is that of the **null hypothesis**:

NULL HYPOTHESIS

A *Null Hypothesis*, denoted H_0 , expresses the **opposite of what a researcher predicts** or anticipates.

This definition may appear confusing at first, but the *general procedure* of hypothesis testing should make its role clear:

1. Define the **null hypothesis** based on the research question.
2. Set a **significance level**. This means choosing a *threshold* to determine the *risk of a false positive error*; that is, the risk of finding a result that *doesn't actually exist*.
3. Calculate the **test statistic**. This consists of computing a value from the sample data using an appropriate test.
4. Determine the **P value**. This is the *probability of observing the data if the null hypothesis were true*.
5. Make a **decision**: we *compare* the p-value with the *significance level* and decide whether we can *reject the null hypothesis* or not.

🔗 HYPOTHESIS TESTING IN SIMPLE TERMS

We formulate H_0 as the **opposite of what we observed**, then we compute the probability of obtaining a **result at least as extreme** as our observation **if H_0 were true** via a *test statistic*.

If the obtained **p-value is small enough** it means that if H_0 were true, our observations would have been very unlikely and we can thus **reject H_0 in favour of our own hypothesis**.

There exist many different **test statistics**, which can be grouped into two main categories:

1. **Parametric tests**: these *assume the data follows a specific distribution* and rely on *population parameters* such as the mean. They are mainly used for *continuous data* (intervals or ratios).
2. **Non parametric tests**: these *do not assume a specific data distribution*. They are applicable to a wider range of data types, including *nominal* and *ordinal*.

Here, we only present two of these tests. The rationale behind these two can then be applied to all the others.

Z TEST

The **Z-Test** can be used to analyse *two populations* and decide whether their **means differ or not** when *variances are known* and the *sample size* is *large enough* (indicatively, we want 30 or more samples).

More precisely, when we use this test we are assessing whether the **observed sample mean is consistent with an hypothesised value of the population mean** under the null hypothesis.

Usually, the *population mean* μ is hypothesised (e.g. from historical data) and the *population standard deviation* σ is assumed to be known.

Under these assumptions, we compute the **z test statistic** as follows:

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Where $\bar{X}$ is the *observed sample mean* and n is the *sample size*.

We then have to compare this z value with a *critical value* that depends on the desired *significance level* and on whether the test is **one-tailed** or **two-tailed**.

🔗 ONE-TAILED VS TWO-TAILED TESTS

- **ONE-TAILED TEST:** We believe that the *population mean is greater than* (in which case we call it a *right-tailed test*) or *lower than* (in which case we call it a *left-tailed test*) the value stated in H_0 .
- **TWO-TAILED TEST:** We simply believe that *the population mean is different from the value stated in H_0* , but we do not care if it's greater or lower.

The most commonly used critical values for the z-test are presented in the following table:

	10%	5%	1%
ONE-TAILED	1.280	1.645	2.330
TWO-TAILED	1.645	1.960	2.580

To better understand how the test works, let's consider the following example.

📋 EXAMPLE

We are told that the **mean score of students** in a class is **70** with a **standard deviation of 10**.

We select a **sample of 50 students** with a **mean score of 80**.

Is there enough evidence to **reject** at a **5% significance level** that the mean score is **70**?

Let's formalise the hypotheses at play for clarity:

- **H_0 (null hypothesis):** the **mean score is 70**.
- **H_1 (our hypothesis, formulated from the sampled students):** the **mean score is greater than 70**.

✍ FORMALITIES ABOUT H0

To be more precise, we should say:

- **H0:** the true (**population**) mean and the **sample mean** are the same.
- **H1:** the **two means are different** and, more precisely, the **sample mean is greater**.

This puts an emphasis on the fact that H0 is stating that *our observations are wrong* (after all, we said H0 is the *opposite* of what we're claiming), but perhaps the (equivalent) formulation presented above is *more intuitive*.

Since we are making an assumption about the "*direction*" of the hypothesis, this is a **one-tailed test**, specifically a *right-tailed test*.

We have $\bar{X} = 80$, $\mu = 70$, $\sigma = 10$ and $n = 50$.

Therefore, by computing z with the above formula we obtain:

$$z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{80 - 70}{10/\sqrt{50}} \approx 7.07$$

Since we want a 5% confidence level, our *critical z value* z^* is:

$$z^* = 1.960$$

🔗 CONCLUSION

We have:

$$z > z^*$$

Therefore we **can reject the null hypothesis** in favour of our own: the two *means are not the same* and the *sample mean is greater*.

The *critical z values* are obtained from a *z value table* (see figure below), which contains the values of the *cumulative distribution function* of a **normal distribution** (remember that this is a function defined as $\text{CDF}(z) = P(Z < z)$ where $Z \sim \mathcal{N}(0, 1)$).

Note that for a 5% confidence level we have to look for a z value corresponding to a 95% probability for a *one-tailed test* and to 97.5% for a *two-tailed test*. This is because in the first case we're interested in the 5% values at **just one end of the distribution**, while in the second case we're interested in the 5% values at **both ends**.

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91309	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997

🔗 ALTERNATIVE Z TEST APPROACH

We can also use this table to *decide if we can reject H_0* by **determining the probability associated with our z value** and **comparing it with the significance level** requested.

In the above example we had $z = 7.07$: note that this value is so extreme that it *doesn't show up in the table* (which only goes up to 3.99).

It's thus reasonable to consider $F(z) = \text{CDF}(7.07) = 0.99999$: this is the *probability that $Z < z$* ; since we are interested in the probability that $Z > z$, we need to consider the complementary value $p = 1 - F(z) = 0.00001$.

We have $p < p^*$, where $p^* = 0.05 = 5\%$ is the *desired significance level*.

NOTE

This approach is perhaps more in line with the explanation of hypothesis testing given at the beginning of the section.

The intuition behind it is that **if H_0 were true, the probability of observing a result at least as extreme** [as the one we're investigating] would be so low that it's **reasonable to reject the null hypothesis** and thus accept our own.

Note that *the higher the confidence level, the smaller our p value needs to be to reject H_0 .*

T TEST

The **t-test** is *very similar to the z-test*, with the key difference that the t-test can be used **when the population variance is not known**.

The *test statistic* is given by:

$$t = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}}$$

ATTENTION

In this case, σ is the **sample standard deviation**, while in the z-test it was the *population standard deviation*.

This test is based on **Student's t distribution**, which is *bell-shaped* and *symmetric*, just like the normal distribution, but has **fatter tails**.

This is to account the fact that we're considering the **sample standard deviation instead of the true population standard deviation**.

As a result, *t distribution tables* have *different values* from those found in a *z table*:

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.90}$	$t_{.95}$	$t_{.99}$	$t_{.995}$	$t_{.9975}$	$t_{.999}$	$t_{.9995}$	$t_{.9999}$	$t_{.99995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

DEGREES OF FREEDOM

Note that the table is organised in rows depending on a parameter called **df**: these are the **degrees of freedom of the distribution**.

In our case, we simply have $df = n - 1$, but interested readers can find more details in > [DETAILS ON DOF](#).

To better understand how this test works, let's walk through an example.

EXAMPLE

Suppose the **hypothesised mean score** of students is **12**.

Suppose we take a sample of students and their marks are:

10.5, 9, 7, 12, 8.5, 7.5, 6.5, 8, 11, 9.5

Is there enough evidence to support that *the two groups* (the sample and the overall student population) have **different means** with a 5% significance level?

We thus have:

- **H0**: the true difference between the groups is zero.

- H1: the true difference is different from zero.

⚠ ATTENTION

We are conducting a **two tailed test** this time: the reason is that we *choose the test before conducting the experiment*.

Since the students are selected *after choosing to use a particular test* and we *don't yet know what their average score is*, we **can only assume the two means differ**, but we *can't yet tell in "which direction" they differ*.

NOTE: Since the test *must be chosen before*, most analyses require a **two tailed test**. We *only use one tailed tests* when we *already have some evidence* (or some kind of information) about the "direction" of the inequality.

The first step is to compute the sample mean and standard deviation.

We obtain $\mu = 8.95$ and $\sigma = 1.8$. Since $n = 10$ we have $df = 9$.

Let's now compute the test statistic:

$$t = \frac{8.95 - 12}{1.80/\sqrt{10}} \approx -5.352$$

With a 5% significance level, a two-tailed test and 9 degrees of freedom, we obtain (from the table) the following critical value t^* :

$$t^* = 2.262$$

Since we have:

$$|-5.352| > 2.262$$

We can **reject the null hypothesis** and say that, indeed, **there is a difference between the sample mean and population mean**.

Note that we could use the absolute value because of the symmetry of the distribution.

CONFIDENCE INTERVALS

We can use **normal distribution** and **Student distribution tables** to **estimate how accurately the sample mean estimates the population mean**.

This is often expressed as a **confidence interval**.

The confidence interval (CI) at a given *confidence level* is calculated as follows:

$$\bar{X} \pm z \frac{\sigma}{\sqrt{n}}$$

$$\bar{X} \pm t \frac{\sigma}{\sqrt{n}}$$

NOTE

The first case refers to *normal distributions*, the second to *Student t distributions*.

In any case, z and t are the values associated with the desired *confidence level*.

Let's take a look at an example:

Characteristic	n	Sample Mean	Standard Deviation (s)
Systolic Blood Pressure	3,534	127.3	19.0
Diastolic Blood Pressure	3,532	74.0	9.9
Total Serum Cholesterol	3,310	200.3	36.8
Weight	3,506	174.4	38.7
Height	3,326	65.957	3.749
Body Mass Index	3,326	28.15	5.32

Assume we have gathered the data in the table above.

We want to generate a **95% confidence interval for systolic blood pressure**.

The z value for 95% is $z = 1.96$. Thus, we obtain:

$$127.3 \pm 1.96 \frac{19}{\sqrt{3534}}$$

$$127.3 \pm 0.63$$

Therefore the **95% confidence interval for systolic blood pressure** is:

$$(126.7, 127.9)$$

MEANING

We are **95% confident** the true mean systolic blood pressure is in the interval above.

This **does not mean** that there is a **95% probability** that the true mean is in the interval; it means that if we can keep repeating the observations over and over, we will **observe a mean inside that interval 95% of the times**.

CORRELATION

Correlation is a **statistical measure** that determines **how closely two variables fluctuate**.

- A *positive correlation* shows the extent to which those variables increase or decrease in parallel.
- A *negative correlation* shows the extent to which one variable increases as the other decreases.

- A *low correlation* value shows that *the two variables are weakly influenced by one another*.

There **isn't one universal way to compute correlation**; in fact, there are many different approaches, each with their *pros and cons*.

Here, we present just two definitions of correlation: **Spearman's** and **Pearson's**.

PEARSON CORRELATION

Pearson correlation measures the **strength of the linear relationship between two random variables**.

📖 PEARSON CORRELATION

Consider two random variables X and Y with standard deviations σ_X and σ_Y . Pearson's correlation coefficient is defined as follows:

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

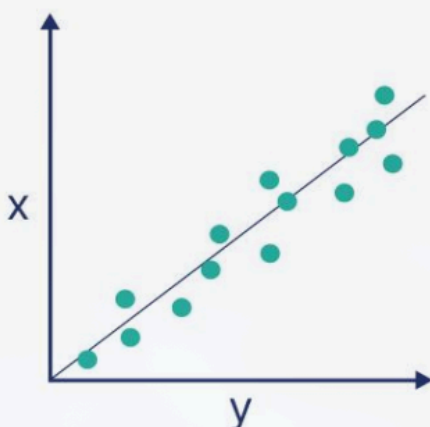
Note that it is essentially a **normalised measurement of the covariance**, such that the result always has a value between -1 and $+1$.

In particular, the **sign** of ρ tells us the **direction of the relationship** between the two variables, while the **absolute value** of ρ tells us **its strength**:

- $0.0 < |\rho| < 0.1$ - *no correlation*.
- $0.1 < |\rho| < 0.3$ - *low correlation*.
- $0.3 < |\rho| < 0.5$ - *medium correlation*.
- $0.5 < |\rho| < 0.7$ - *high correlation*.
- $0.7 < |\rho| < 1.0$ - *very high correlation*.

Strong positive correlation

$$r > .5$$



Strong negative correlation

$$r < -.5$$



🔍 OBSERVATION

A low Pearson correlation value does **not necessarily mean** the two variables aren't correlated.

As stated above, ρ measure the **strength** of the **linear relationship** between the two: if the variables are correlated, but their **relationship is non-linear**, then Pearson's coefficient **can't detect it**.

For discrete random variables (which cover most of the applications we're interested in), we can rewrite ρ in terms of the *observed values* x_i and y_i and the *mean values* $\bar{x}$ and $\bar{y}$:

$$\rho = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$$

Let's now look at an example. Consider the scores of 7 students in *maths* and *science*:

STUDENT	MATHS (X)	SCIENCE (Y)
1	98	88
2	87	92
3	91	95
4	75	82
5	81	74
6	68	78
7	93	96

We have $\bar{x} = 84.71$ and $\bar{y} = 86.43$.

We now compute the values to find ρ :

STUDENT	$x_i - \bar{x}$	$y_i - \bar{y}$	$(x_i - \bar{x})(y_i - \bar{y})$	$(x_i - \bar{x})^2$	$(y_i - \bar{y})^2$
1	13.29	1.57	20.87	176.62	2.46
2	2.29	5.57	12.76	5.24	31.02
3	6.29	8.57	53.91	39.56	73.44
4	-9.71	-4.43	43.02	94.28	19.62
5	-3.71	-12.43	46.12	13.76	154.5
6	-16.71	-8.43	140.87	279.22	71.06
7	8.29	9.57	79.34	68.72	91.58

And thus we find:

$$\rho = 0.724$$

The performances in mathematics and science are **positively (highly) correlated**.

REPORTING THE CORRELATION VALUE

When we **report a correlation value** we also need to accompany it with information regarding the **statistical significance** of our observation.

The general form of this report consists in **stating the null hypothesis** and **rejecting it with some statistical significance**.

Usually, **H₀** is stated as follows:

H₀: There is **no linear association between the variables** in the population.

We can *decide whether to reject H₀ or not* by using a **t test**, which in the case of Pearson's correlation is given by:

$$t = \rho \frac{\sqrt{n-2}}{\sqrt{1-\rho^2}}$$

NOTE

1. A *p*-value can be obtained from a **t distribution table** with $df = n - 2$.
Intuitively, we're *losing two degrees of freedom* because we are *estimating two values* (the *mean for each variable*) before calculating ρ .
2. By rewriting the above formula in the following way, we immediately recognise it has the **same form as the t statistic** seen in the relevant section:

$$t = \frac{\rho - 0}{\sqrt{1-\rho^2}/\sqrt{n-2}}$$

Where:

- $\rho = 0$ is the **value of the correlation according to H₀** (which claims *there is no correlation*).
- $\sigma = \sqrt{1-\rho^2}$ is the **fraction of the standard deviation that is not explained by the correlation**.

CLARIFICATION

It is important to realise that **statistical significance does not indicate the strength of Pearson's correlation**.

In fact, achieving a value of $p = 0.001$, for example, does *not mean that the relationship is stronger* than if we had achieved a value of $p = 0.04$.

This is because the significance test is **only investigating whether we can reject or fail to reject the null hypothesis**.

SPEARMAN CORRELATION

We've seen how Pearson's correlation is *limited to linear relationships*.

Spearman's correlation, also known as **Spearman's rank coefficient**, *overcomes this limitation* by evaluating **how well** the correlation between two variables **can be formulated by a monotonic function**.

To achieve this, Spearman's correlation uses **data rank** to measure monotonicity between **ordinal or continuous variables**.

SPEARMAN'S RANK COEFFICIENT

Consider two random variables X and Y , with standard deviations σ_X and σ_Y . Spearman's rank coefficient is defined as the **Pearson correlation between the rank variables**:

$$r_s = \rho[R[X], R[Y]] = \frac{\text{cov}[R[X], R[Y]]}{\sigma_{R[X]}\sigma_{R[Y]}}$$

Where $d_i := R[X_i] - R[Y_i]$ is the *difference between the two ranks of each observation*.

SIMPLIFICATION

Only when all n **ranks are distinct integers** (there are no ties), the coefficient can be computed using the popular (simpler) formula:

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2 - 1)}$$

Note that r_s can take values in the range $[-1, +1]$ where:

- A value of **+1** means there is a *perfect association of rank*. This means the *correlation* of the two variables can be *perfectly described by a monotonic (increasing) function*.
- A value of **0** means *there is no association between ranks*.
- A value of **-1** means *perfect negative association of rank*. This means the *correlation* of the two variables can be *perfectly described by a monotonic (decreasing) function*.

Once again, let's look at example to better understand how to calculate r_s .

Consider the scores of 5 students in maths and science, as reported by the table

below:

STUDENT	MATHS (X)	SCIENCE (Y)
A	35	24
B	20	35
C	49	39
D	44	48
E	30	45

We need to *compute the rank of each observation*: the highest mark will get a rank of 1, the lowest will get 0:

STUDENT	MATHS (X)	$R[x_i]$	SCIENCE (Y)	$R[y_i]$	d_i	d_i^2
A	35	3	24	5	2	4
B	20	5	35	4	1	1
C	49	1	39	3	2	4
D	44	2	48	1	1	1
E	30	4	45	2	2	4

Notice that there are *no tied scores*, which means we can use the simplified version of the formula.

We thus obtain:

$$r_s = 1 - \frac{6 \cdot 14}{5(25 - 1)} = 0.3$$

Since the value isn't particularly high, we conclude that there's a **weak correlation** between the two ranks.

Just like before, we also need to provide information regarding the **statistical significance of our observation**.

To do so, we need to *set the desired significance level* and then *check whether our computed Spearman correlation coefficient is high enough*.

To check this, we use the **Spearman critical value table** below:

	Nominal α					
n	0.10	0.05	0.025	0.01	0.005	0.001
4	1.000	1.000	-	-	-	-
5	0.800	0.900	1.000	1.000	-	-
6	0.657	0.829	0.886	0.943	1.000	-
7	0.571	0.714	0.786	0.893	0.929	1.000
8	0.524	0.643	0.738	0.833	0.881	0.952
9	0.483	0.600	0.700	0.783	0.833	0.917
10	0.455	0.564	0.648	0.745	0.794	0.879
11	0.427	0.536	0.618	0.709	0.755	0.845
12	0.406	0.503	0.587	0.678	0.727	0.818
13	0.385	0.484	0.560	0.648	0.703	0.791
14	0.367	0.464	0.538	0.626	0.679	0.771
15	0.354	0.446	0.521	0.604	0.654	0.750
16	0.341	0.429	0.503	0.582	0.635	0.729
17	0.328	0.414	0.488	0.566	0.618	0.711
18	0.317	0.401	0.472	0.550	0.600	0.692
19	0.309	0.391	0.460	0.535	0.584	0.675
20	0.299	0.380	0.447	0.522	0.570	0.662
21	0.292	0.370	0.436	0.509	0.556	0.647
22	0.284	0.361	0.425	0.497	0.544	0.633
23	0.278	0.353	0.416	0.486	0.532	0.621
24	0.271	0.344	0.407	0.476	0.521	0.609
25	0.265	0.337	0.398	0.466	0.511	0.597
26	0.259	0.331	0.390	0.457	0.501	0.586
27	0.255	0.324	0.383	0.449	0.492	0.576
28	0.250	0.318	0.375	0.441	0.483	0.567
29	0.245	0.312	0.368	0.433	0.475	0.558

NOTE

1. To get the critical value we need to check the **row corresponding** to $n - 2$.
2. The **nominal alpha** is referred to a **one tailed test**. If we're performing a *two tailed test* we need to consider $\alpha/2$.

Note that, *in most cases*, we **use a two-tailed test** for correlation.

This is because we are simply asking: *is there a relationship between these two variables?* We **don't mind** if the correlation is positive or negative.

However, we use a **one-tailed test** when we have **prior information** (or a specific *prediction* or *theory*) about the **direction of the relationship**.

For example, we could predict that more hours of study will increase exam scores. In this case, we are only looking for a positive correlation; we do not care about a negative one.

PEARSON AND SPEARMAN COMPARISON

Pearson correlation is particularly good at **capturing linear relationships** between variables.

Its sensitivity to linear patterns makes it a powerful tool when investigating relationships governed by a *consistent linear trend*. Moreover, the standardised nature of the coefficient allows for easy comparison across different datasets.

However, it's crucial to note that Pearson is **susceptible to the influence of outliers**. If a dataset contains **extreme values** they can **impact the calculation**, leading to **inaccurate interpretations**.

Pearson correlation **effectiveness diminishes** when faced with **curvilinear patterns**. This limitation arises from Pearson's **inherent assumption of linearity**, making it ill-suited to capture the nuances of non-linear relationships.

Unlike Pearson, **Spearman does not assume linearity** and is **robust to outliers**. It focuses on the ordinal nature of data, making it a valuable tool when the relationship between variables is **more about the order than the specific values**. In fact, Spearman **assesses monotonicity**, which can be linear, but also non-linear.

OTHER APPROACHES TO CORRELATION

Notice that there are **other correlation approaches** proposed in the literature, such as **Kendall Tau** (or simply *Kendall's correlation*), which is a *non parametric alternative* to Pearson's correlation.

Since it also identifies monotonic relationships between the variables, it is **commonly considered as an alternative to Spearman's correlation**.

Spearman is preferred for *weak correlations in small datasets*; in **large samples**, **Kendall Tau's reliability surpasses Spearman**.

Kendall's Tau is a robust estimator against outliers and *non-normality* (non-normal distributions).

DETAILS ON DOF

Estimates of statistical parameters can be **based upon different amounts of information** about the data.

DEGREES OF FREEDOM (STATISTICS)

Degrees of freedom are the **number of independent pieces of information** that go into the **estimate of a parameter**.

In general, the degrees of freedom of an estimate of a parameter are equal to the **number of independent observations** that go into the estimate **minus the number of parameters used as intermediate steps** in the estimation of the parameter itself.

For example, if the variance is to be estimated from a random sample of n independent scores, then the degrees of freedom are $n - 1$ (number of independent scores, n , minus the number of parameters estimated as intermediate steps, in this case, the variance).

When computing the **Pearson's correlation**, we are **estimating two parameters**: the *mean for each variable*.

🔗 OBSERVATION

After all, we said that Pearson's correlation measures the *strength of a linear relationship between the variables*.

Now, *how many parameters do we need to completely characterise a linear relationship?* The answer is **two**: a **slope** and an **intercept**.

In fact, computing Pearson's correlation is *equivalent* to fitting the model:

$$Y = \beta_0 + \beta_1 X$$

We **start with n data points**, then **two parameters are estimated**, and thus the residual degrees of freedom are $n - 2$.

Mathematically, degrees of freedom can also be interpreted as the **number of dimensions of the domain of a random vector**, or essentially the *number of "free" components* (how many components need to be known before the vector is fully determined).

To better understand this interpretation, consider the following process:

- Let's assume we made n *independent measurements*. We can represent this *set of observations* with a (*random*) vector $v = (x_1, \dots, x_n) \in \mathbb{R}^n$, where **each component x_i is the i -th observation**.
At this point, since we have **no constraints**, all n components can vary freely.
- We then **compute the mean** of the sample. This means **adding a constraint** of the form $x_1 + x_2 + \dots + x_n = n\bar{x}$: an *equation that forces one of the components*.
The **random vector** can now **only vary in a subspace of the original**; since there is just one constraint, we're effectively forcing the vector into $\mathbb{R}^{n-1}$.
- Similarly, every time we *estimate a parameter* we are **adding a constraint** that **further reduces the dimension of the subspace** the random vector is *forced to be in*.

Now let's see how this interpretation works in the case of **correlation**.

Now, **each observation is a pair of values** (x_i, y_i) , and since we made n observations, we have $2n$ **total values**.

This means that the vector representing the observations has $2n$ components.

Furthermore, we know that to calculate the correlation between X and Y we **need to estimate 2 parameters**.

Thus, we are lead to think that the *leftover degrees of freedom* should be $2n - 2$; but wait, we have seen that, in reality, the degrees of freedom are $n - 2$.

How do we explain this discrepancy?

🔑 KEY INTUITION

The answer lies in the fact that **assuming the two variables are correlated** translates to constraints of the form:

$$Y = f(X) \implies y_i = f(x_i), i = 1, 2, \dots, n$$

This means we **actually have $n+2$ constraints**: n coming from the *assumption of correlation* and 2 being the *estimated means*.

Thus, we finally obtain $df = 2n - (n + 2) = n - 2$.

SURVIVAL ANALYSIS

We are often interested in *assessing whether there are differences in survival (or cumulative incidence of event) among different groups of participants in a statistical study*. For example, in a clinical trial with a survival outcome, we might be interested in *comparing survival between participants receiving the drug chosen by the neural network A as compared to the drug chosen by the neural network B*.

📖 SURVIVAL ANALYSIS

Survival analysis is a branch of statistics for **analysing the expected duration of time until one event occurs**, such as death in biological organisms or failure in mechanical systems.

This topic is often called **reliability theory, reliability analysis** engineering.

In this section, topics will be presented from the perspective of the *analysis of biological systems* for the sake of intuition, but the same concepts can be applied to many other fields just by adopting a **different definition** of "**experiencing an event**", as seen in the definition above.

Before diving in, we present the *context* of this kind of studies and the *problems we're facing* when conducting them.

For a start, **complete data** (*actual time to event data*) is **not always available** on each participant in a study.

In many studies, *participants are enrolled over a period of time* (months or years) and the study ends on a specific calendar date. Thus, participants who **enrol later are followed for a shorter period** than participants who enrol early. Some participants may also *drop out of the study before the end of the follow-up period* (e.g., move away, become disinterested) and others may *die during the follow-up period* (assuming the outcome of interest is not death).

In each of these instances, we have **incomplete follow-up information**.

True survival time (sometimes called failure time) is not known because the study ends or because a participant drops out of the study before experiencing the event. What we know is that the participants' survival time is greater than their last observed follow-up time. These times are called **censored times**.

In general, there are **different types of censoring**.

The most common is called **right censoring** and occurs when a **participant does not have the event of interest during the study** and thus their last observed follow-up time is less than their time to event.

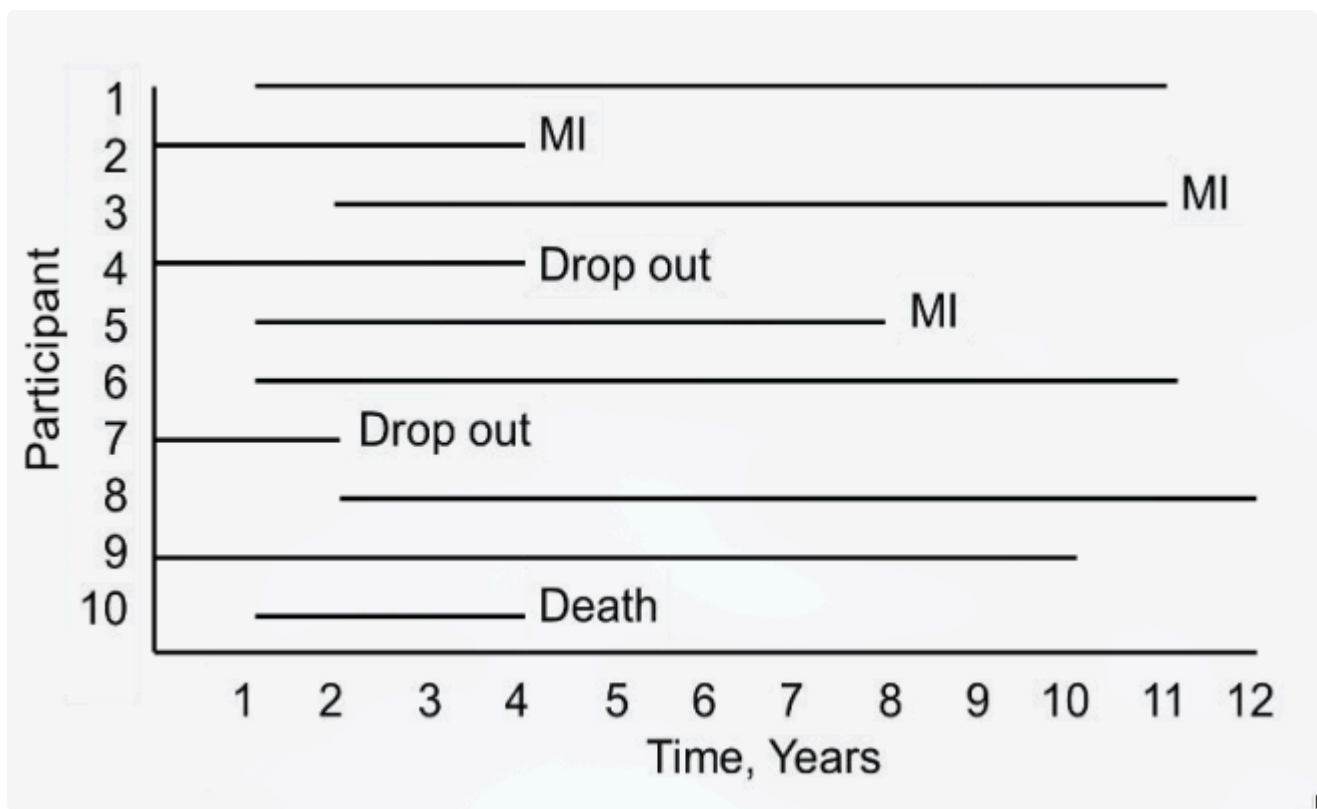
This can occur when a *participant drops out of the study before it ends* or when a *participant is event free at the end of the observation period*.

Consider the following example:

EXAMPLE

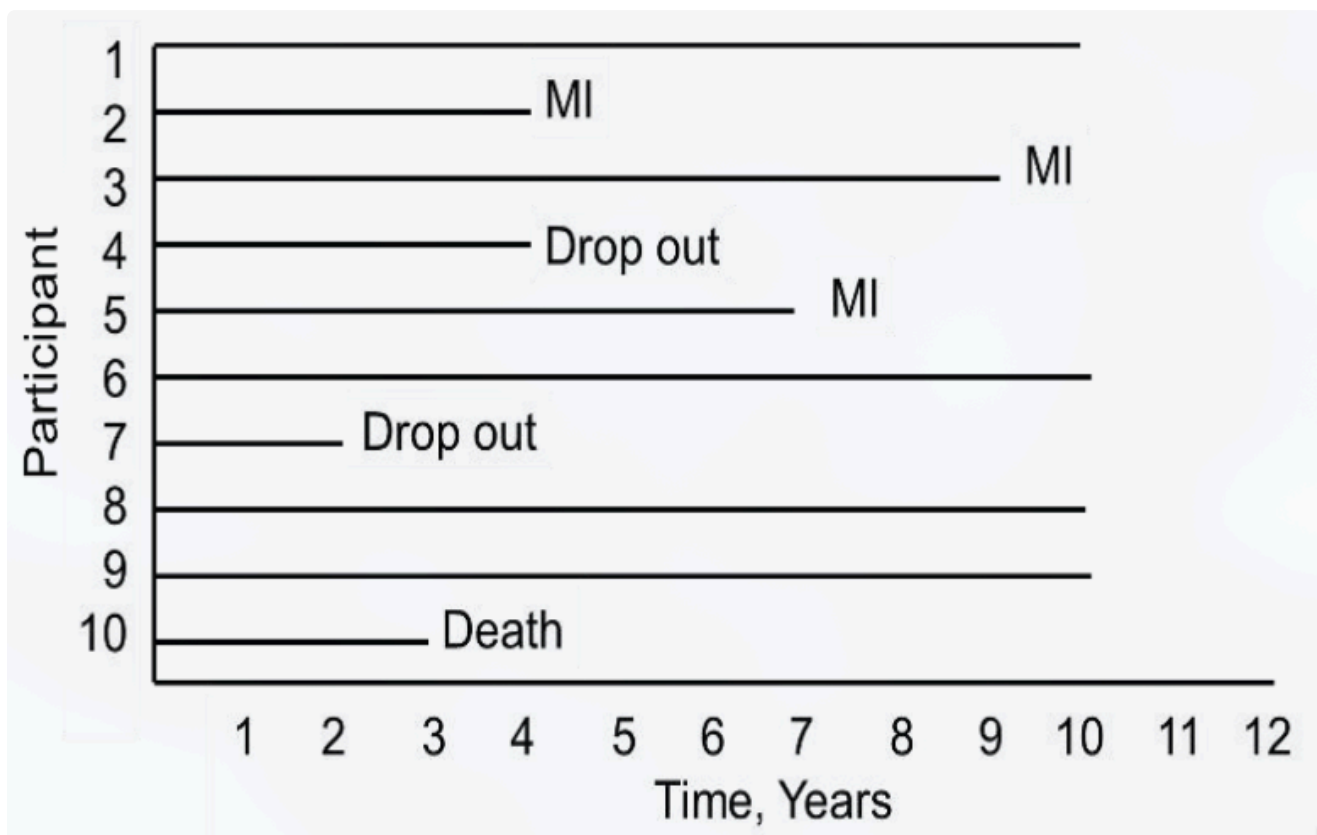
Participants are **recruited** into the study **over a period of two years** and are **followed for up to 10 years**.

The figure below indicates *when they enrolled and what subsequently happened to them* during the observation period.



During the study period, *three participants suffer myocardial infarction (MI), one dies, two drop out of the study (for unknown reasons), and four complete the 10-year follow-up without suffering MI.*

The next figure shows the same data, but survival times are now starting at a common "zero time" (i.e., as if all participants enrolled in the study at the same time).



Based on this data, we may be interested in calculating **what is the likelihood that a participant will suffer an MI over 10 years.**

Three of the 10 participants **suffer MI** over the course of follow-up, so we are lead to think that an **estimate for the likelihood** is 30%.

🔍 OBSERVATIONS

1. The problem is that this is **probably an underestimate** of the true likelihood as **two participants dropped out** and might have suffered an MI had they been observed for the full 10 years. Their **observed times are censored**.
2. Furthermore, **one participant dies** after 3 years of follow-up, raising the question of whether we should include it in the analysis or not.
3. **If we included these 3 participants, how should we proceed** with the statistical analysis?

Notice that if we *exclude all three participants who haven't completed the follow up study*, the estimate of the likelihood that a participant suffers an MI is $3/7 = 43\%$, **substantially higher than the initial estimate** of 30%.

The fact that all participants are often not observed over the entire follow-up period makes survival data unique.

In this small example, participant 4 is observed for 4 years and over that period does not have an MI. Participant 7 is observed for 2 years and over that period does not have an MI. While they do not suffer the event of interest, they still **contribute important information** that **should be accounted for**.

Survival analysis techniques are aimed at making use of this information in the estimate of the probability of event.

✍ NOTE

There are a number of **popular parametric methods** that are used to model survival data, and they differ in terms of the **assumptions** that are made about the **distribution of survival times** in the population.

In this section, we **focus on non parametric methods**, which make **no assumptions about how the probability** that a person experiences the event **changes over time**.

KAPLAN-MEIER SURVIVAL CURVE

We now present one possible *non parametric method* for the study of survival analysis problems, know as **Kaplan-Meier approach**.

Let's consider the following example:

EXAMPLE

A study involves **20 participants** who are 65 years of age and older. They are **enrolled over a 5 year period** and are **followed for up to 24 years**, until they die, the study ends, or they drop out of the study (lost to follow-up).

The study is summarised in the following table:

PARTICIPANT ID	YEAR OF DEATH	YEAR OF LAST CONTACT
1		24
2	3	
3		11
4		19
5		24
6		13
7	14	
8		2
9		18
10		17
11		24
12		21
13		12
14	1	
15		10
16	23	
17		6
18	5	
19		9
20	17	

KEY ASSUMPTION

The key assumption the approach is based on is that **censoring is independent of the likelihood of experiencing the event** of interest and that **survival probabilities are comparable in participants** who are recruited early and later into the study.

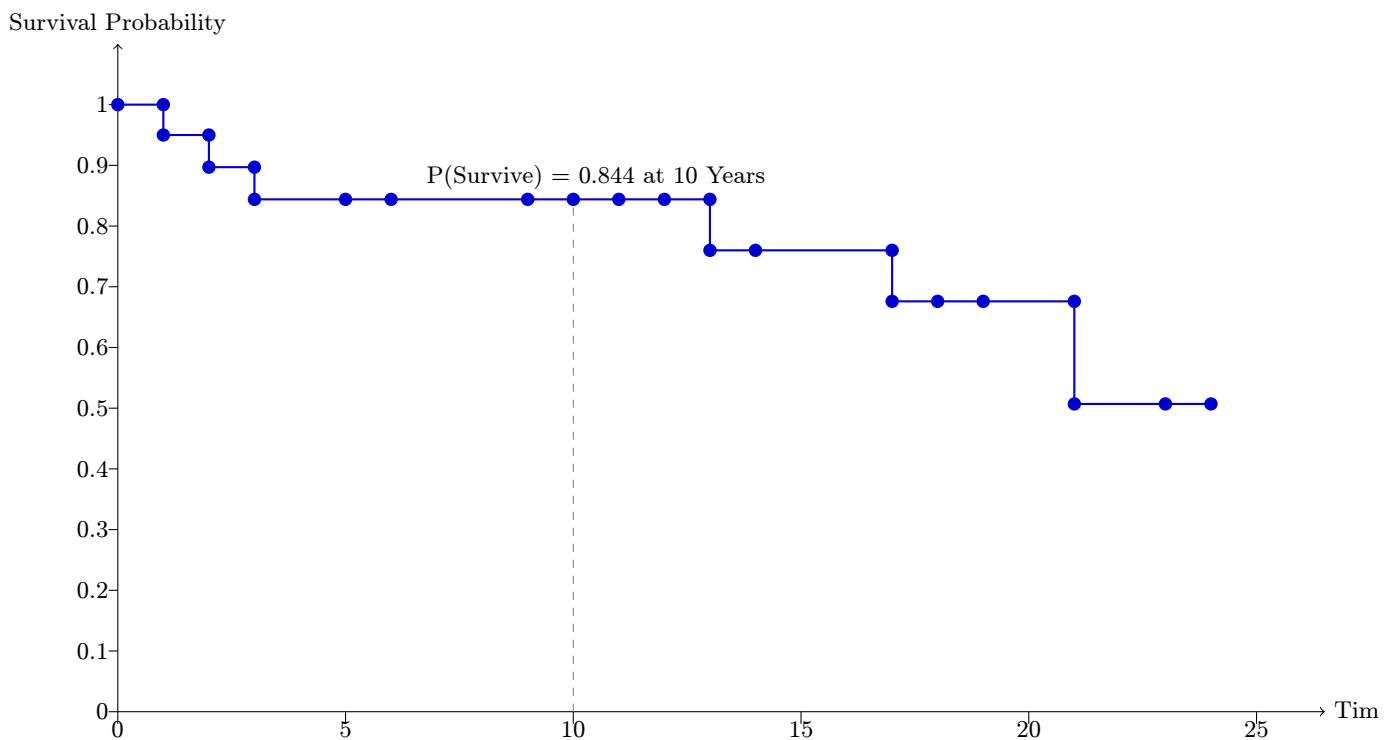
We organise the data in a table with **one row for each year of the follow up period**; at each year, we use the **number of participants at risk** N_t , the **number of deaths** D_t and the **number of censored participants** C_t to compute the survival probability S_t :

$$S_t = S_{t-1} \cdot \left(\frac{N_t - D_t}{N_t} \right), t \geq 1$$

The resulting table is shown below:

YEAR	N_t	D_t	C_t	S_t
0	20			1
1	20	1		$1 \cdot \frac{20-1}{20} = 0.950$
2	19		1	$0.950 \cdot \frac{(19-0)}{19} = 0.950$
3	18	1		0.897
5	17	1		0.844
6	16		1	0.844
9	15		1	0.844
10	14		1	0.844
11	13		1	0.844
12	12		1	0.844
13	11		1	0.844
14	10	1		0.760
17	9	1	1	0.676
18	7		1	0.676
19	6		1	0.676
21	5		1	0.676
23	4	1		0.507
24	3		3	0.507

We can plot the data from this table to obtain the **survival curve** shown in the figure below:



Note that some investigators prefer to generate **cumulative incidence curves**, as opposed to survival curves, which show the **cumulative probabilities of experiencing the event** of interest.

Cumulative incidence I_t , or *cumulative failure probability*, is computed as the *complementary of the survival probability*:

$$I_t = 1 - S_t$$

From the *survival curve* we see that the proportion of participants surviving past 10 years is 84%, and the proportion of participants surviving past 20 years is 68%.

The **median survival** is estimated by locating 0.5 on the Y axis and reading over and down to the X axis: the median survival is approximately 23 years.

The *median survival* is commonly reported in studies where different drugs are compared.

NOTE

The estimates of *survival probabilities at specific times* and the *median survival time* are **point estimates**.

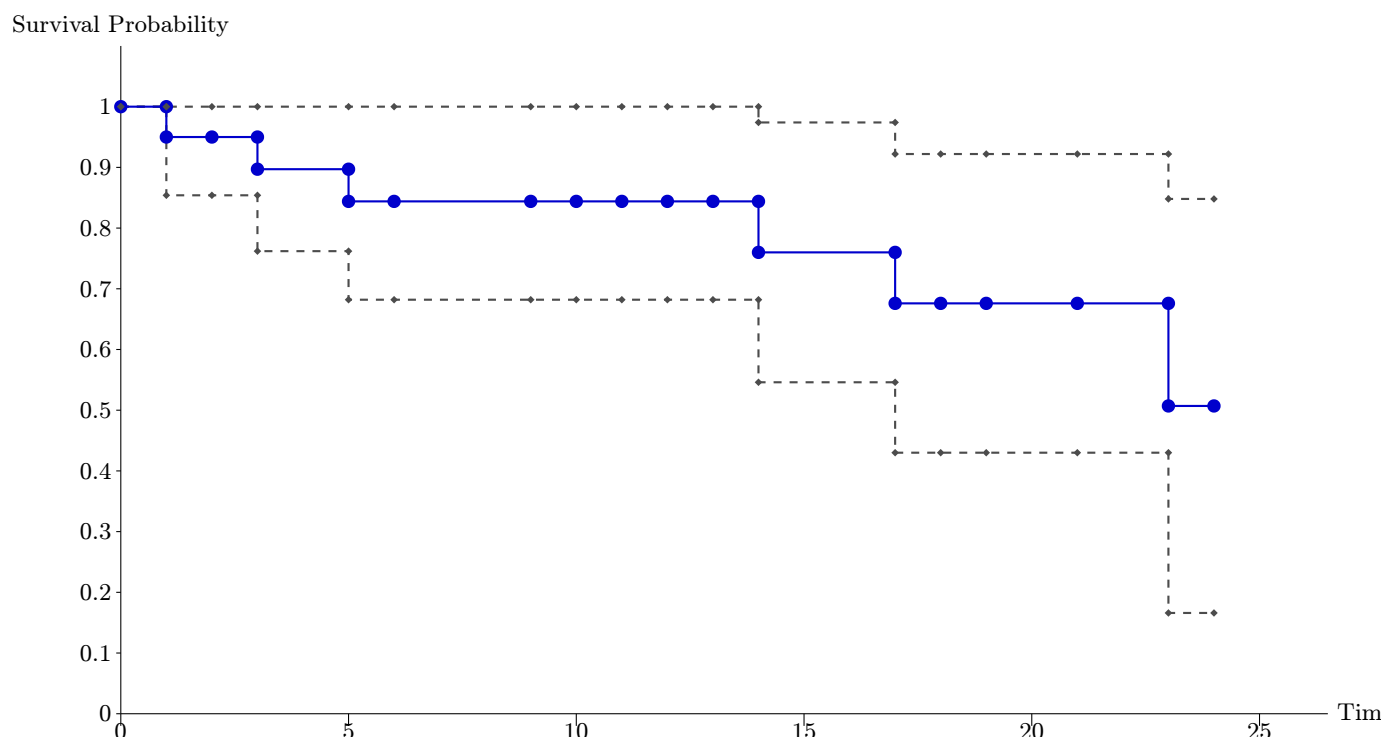
It is possible to produce **standard errors** and **confidence intervals** of survival probabilities.

A popular way to estimate the **standard error** of the **survival estimates** is given by *Greenwood's formula*:

$$SE(S_t) = S_t \sqrt{\sum_{t=1}^{t_{max}} \frac{D_t}{N_t(N_t - D_t)}}$$

The standard errors at each time can be used to generate **upper and lower bounds of the confidence interval**.

In the figure below, dotted lines represent the *95% confidence limits* for the example discussed above. These bounds are given by $z^* \cdot \text{SE}(S_t)$ with $z^* = 1.96$.



LOG RANK TEST

The **log rank test** is a popular test to *test the null hypothesis of no difference in survival between two or more independent groups*.

The test compares the entire survival experience between groups and can be thought of as a test of **whether the survival curves are identical (overlapping) or not**.

Let's discuss a case with 2 curves: S_{1t} and S_{2t} , and suppose we want *statistical significance* $\alpha = 0.05$.

The hypotheses at play are:

- **H0:** The **two survival curves are identical**.
- **H1:** The two survival curves are **different**.

The log rank statistic is **approximately distributed as a chi-square test statistic**.

There are *several forms of the test statistic*, and they vary in terms of how they are computed. We use the following:

$$\chi^2 = \sum_j \frac{(\sum_t O_{jt} - \sum_t E_{jt})^2}{\sum_t E_{jt}}$$

Where $\sum_t O_{jt}$ represents the **sum of the observed number of events in the j-th group over time** and $\sum_t E_{jt}$ represents the **sum of the expected number of events in the j-th group over time**.

To see the test in action, let's apply it to the following case of study:

EXAMPLE

A small clinical trial is run to **compare two combinations of treatments** in patients with advanced gastric cancer.

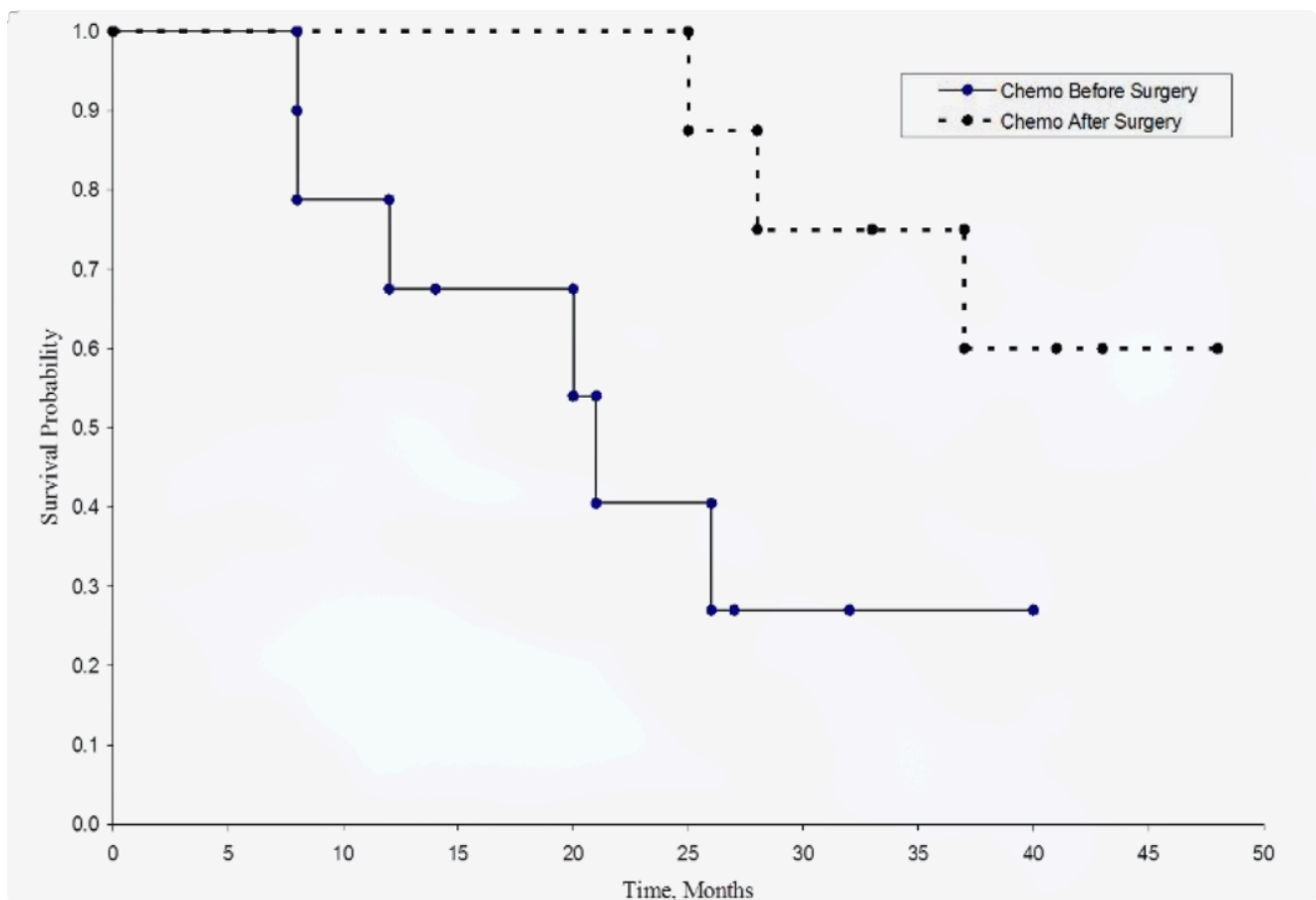
Twenty participants with stage IV gastric cancer who consent to participate in the trial are randomly assigned to be treated with **protocol A** or **protocol B**.

Six participants in the *chemotherapy before surgery* group die over the course of follow-up as compared to three participants in the *chemotherapy after surgery* group. Other participants in each group are *followed for varying numbers of months*, some to the end of the study at 48 months (in the chemotherapy after surgery group).

Using the procedures outlined earlier, we begin applying the **Kaplan-Meier approach**.

Chemotherapy Before Surgery		Chemotherapy After Surgery	
Month of Death	Month of Last Contact	Month of Death	Month of Last Contact
8	8	33	48
12	32	28	48
26	20	41	25
14	40		37
21			48
27			25
			43

The resulting survival curves are shown below:



🔍 OBSERVATION

The survival probabilities for the *chemotherapy after surgery* group are **higher** than the survival probabilities for the *chemotherapy before surgery* group, **suggesting a survival benefit**.

However, these survival curves are **estimated from small samples**.

Now we proceed to compute the χ^2 value. To do so, we need the **observed events** and the **expected number of events at each time step**; we know the former, but lack the latter.

To obtain these, we need to calculate what the **expected number of events** would be **if the null hypothesis were true**.

🔍 EXPECTED NUMBER OF EVENTS

Since H_0 states that *all groups have the same survival experience*, we **expect the events** to be **uniformly distributed in the two groups**. That is:

$$E_{1t} = O_t \frac{N_{1t}}{N_t}, E_{2t} = O_t \frac{N_{2t}}{N_t}$$

Furthermore, the **total number of (observed) events is fixed**. This means that (assuming we're dealing with k groups), once we know **differences** (between

observed and expected events) for $k - 1$ groups, the **difference for the last group is also already determined**, since they must sum to zero.

As a consequence, the **degrees of freedom** are $k - 1$: this is important because the computed χ^2 value must be *compared against a critical value* obtained from a χ^2 distribution table, with values that *depend on the degrees of freedom* of the distribution itself.

We thus obtain the following table:

MONTH	N_{1t}	N_{2t}	TOTAL N_t	O_{1t}	O_{2t}	TOTAL O_t	E_{1t}	E_{2t}
8	10	10	20	1	0	1	0.500	0.500
12	8	10	18	1	0	1	0.444	0.556
14	7	10	17	1	0	1	0.412	0.588
21	5	10	15	1	0	1	0.333	0.667
26	4	8	12	1	0	1	0.333	0.667
27	3	8	11	1	0	1	0.273	0.727
28	2	8	10	0	1	1	0.200	0.800
33	1	7	8	0	1	1	0.125	0.875
41	0	5	5	0	1	1	0.000	1.000
ROW TOTAL:				6	3		2.620	6.380

We can finally **calculate the test statistic**:

$$\chi^2 = \frac{(6 - 2.620)^2}{2.620} + \frac{(3 - 6.380)^2}{6.380} \approx 6.151$$

Notice that in this example we have $k = 2$, therefore $k - 1 = 1$ degree of freedom.

By looking at a *critical value table* for the distribution (these can be found online or in textbooks) we obtain $\chi_*^2 = 3.84$.

Since **we observe** $\chi^2 = 6.151 > \chi_*^2 = 3.84$ **we can reject the null hypothesis**: we have **significant evidence to show that the two curves are different**.

NOTE

Notice that it is a **goodness-of-fit test**: it compares observed data with the expected data and *determines if the difference is due to chance*.

HAZARD AND HAZARD RATIO

In the context of survival analysis, an important indicator that needs to be reported is the **hazard** H of the group.

HAZARD

It represents the **expected number of events per one unit of time**.

It's a very intuitive metric: if the hazard of a group at time t is 0.2 and t is measured in months, it means that *on average*, 0.2 events are **expected per person per month**. Another interpretation is based on the *reciprocal of the hazard*. In the example above we have $\frac{1}{H} = \frac{1}{0.2} = 5$: this means the **expected event-free time** is 5 months per person at risk.

In most situations, we are interested in **comparing groups with respect to their hazards**.

The hazard ratio is the ratio of the total number of observed to expected events in two independent comparison groups:

HAZARD RATE

Given two groups A and B , their **hazard ratio** is defined as:

$$HR = \frac{\sum_t O_{At} / \sum_t E_{At}}{\sum_t O_{Bt} / \sum_t E_{Bt}}$$

Where O_{Gt} is the number of *observed events in group G at time t* and E_{Gt} is the *number of expected events in group G at time t* .

We can calculate the *hazard ratio* from the previous example:

$$HR = \frac{\frac{O_t}{E_t} \Big|_{\text{Chemo before}}}{\frac{O_t}{E_t} \Big|_{\text{Chemo after}}} = 4.870$$

Thus, the **risk of death** is 4.87 times **higher** in the chemotherapy before surgery group as compared to the chemotherapy after surgery group.

COMPARING CLASSIFIERS

An important application of hypothesis testing in machine learning is **comparing the performances of classifiers**.

The goal is to determine whether **different classifiers exhibit different performances** and, in such case, **how large are these differences**.

In the following sections, we present some of the methods used to *compare performances* of classifiers and, in general, other machine learning models.

MCNEMAR'S TEST STATISTIC

McNemar's Test (sometimes also called "*within-subjects chi-squared test*", because the test statistic follows a χ^2 distribution) is a **statistical test for paired nominal data**. In the context of machine learning (or statistical) models, we can use McNemar's test to **compare the predictive accuracy of two models**.

Given two models, we consider the following table:

	MODEL 2 CORRECT	MODEL 2 WRONG
MODEL 1 CORRECT	a	b
MODEL 1 WRONG	c	d

Given the table above, the **hypotheses** at play are formulated as follows:

- **H0:** the probabilities $p(b)$ and $p(c)$ are **the same**: none of the two models performs better than the other.
- **H1:** the **two models perform differently**, thus $p(b) \neq p(c)$.

The test statistic follows a **chi squared distribution** with **one degree of freedom** and is computed as follows:

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c}$$

Consider the following example:

	MODEL 2 CORRECT	MODEL 2 WRONG
MODEL 1 CORRECT	9945	15
MODEL 1 WRONG	25	15

Can we state that the **two models perform differently**, with $\alpha = 0.05$ statistical significance?

By computing the test statistic we obtain $\chi^2 = 2.025$, which corresponds to a p -value of 0.15. Since this is *larger than the significance threshold*, we **cannot reject the null hypothesis**: there isn't enough evidence to suggest that the two models have different performances.

NOTE

An **exact binomial test** is **recommended for small sample sizes** ($b + c < 25$), since the *test statistic* may *not be well-approximated* by the chi-squared distribution.

The standard McNemar test assumes continuous data, but in the case of small samples, the possible values of the statistical test are **few and "jump"** from one number to another, thus making the approximation *too optimistic*.

The exact p -value can be computed as follows:

$$p = 2 \sum_{i=b}^n \binom{n}{i} 0.5^i (1 - 0.5)^{n-i} = 2(0.5^n) \sum_{i=b}^n \binom{n}{i}$$

Where $n = b + c$.

WILCOXON SIGNED RANK TEST

The **Wilcoxon signed-ranks test** is a **non-parametric test** which ranks the **differences in performances of two classifiers** for each data set.

Let's see how it works.

Let d_i be the **difference between the performance scores** of the two classifiers on the i -th dataset out of the N total.

These differences are **ranked according to their absolute values**, with *average ranks* being assigned *in case of ties*.

Now let R^+ be the sum of ranks for the data sets on which the **second model outperformed the first**, and, similarly, let R^- be the sum of ranks for the data sets on which *the first model outperformed the first*:

$$R^+ = \sum_{d_i > 0} \text{rank}(d_i) + \frac{1}{2} \sum_{d_i = 0} \text{rank}(d_i)$$

$$R^- = \sum_{d_i < 0} \text{rank}(d_i) + \frac{1}{2} \sum_{d_i = 0} \text{rank}(d_i)$$

🔍 OBSERVATION

Notice that ranks of $d_i = 0$ are *split evenly across the two* and if there's an odd number of them, one is ignored

Now let T be the **smaller of the two sums**:

$$T := \min\{R^+, R^-\}$$

This value is then compared to a **critical value** T^* , obtained as explained below, to determine whether we can reject or not the null hypothesis.

Notice that, just like earlier, the hypotheses at play are:

- **H0**: the two classifiers have **identical performances**.
- **H1**: the two classifiers have **different performances**.

Most books include a **table of critical values** for T for N up to 25 (see table in the example below).

Otherwise, for larger numbers of data sets, we can compute a **test statistic** as follows:

📖 WILCOXON TEST STATISTIC

$$z = \frac{T - \frac{1}{4}N(N+1)}{\sqrt{\frac{1}{24}N(N+1)(2N+1)}}$$

This test statistic is *approximately normally distributed* and we can thus use z -value tables.

Let's take a look at an example.

EXAMPLE

Two different **classifiers** (C4.5 and C4.5+m) are used to make *medical predictions* on patients with *prior conditions*.

The performance of the two classifiers are reported in the following table, along with the ranks of each difference.

datasets:	C4.5	C4.5+m	difference	rank
adult (sample)	0.763	0.768	+0.005	3.5
breast cancer	0.599	0.591	−0.008	7
breast cancer wisconsin	0.954	0.971	+0.017	9
cmc	0.628	0.661	+0.033	12
ionosphere	0.882	0.888	+0.006	5
iris	0.936	0.931	−0.005	3.5
liver disorders	0.661	0.668	+0.007	6
lung cancer	0.583	0.583	0.000	1.5
lymphography	0.775	0.838	+0.063	14
mushroom	1.000	1.000	0.000	1.5
primary tumor	0.940	0.962	+0.022	11
rheum	0.619	0.666	+0.047	13
voting	0.972	0.981	+0.009	8
wine	0.957	0.978	+0.021	10

OBSERVATIONS

Notice that, as stated above, **average ranks are assigned in case of ties**. Here, for example, a difference of 0.000 appears in 2 data sets: instead of assigning ranks 1 and 2, we assign their average (1.5).

From the table above, we compute:

- $R^+ = 3.5 + 9 + 12 + 5 + 6 + 14 + 11 + 13 + 8 + 10 + 1.5 = 93$
- $R^- = 7 + 3.5 + 1.5 = 12$

Therefore we have $T = R^- = 12$.

Two-Sided Test α	0.1	0.05	0.02	0.01
One-Sided Test α	0.05	0.025	,01	0.005
n				
5	1			
6	2	1		
7	4	2	0	
8	6	4	2	0
9	8	6	3	2
10	11	8	5	3
11	14	11	7	5
12	17	14	10	7
13	21	17	13	10
14	26	21	16	13
15	30	25	20	16
16	36	30	24	19
17	41	35	28	23
18	47	40	33	28
19	54	46	38	32
20	60	52	43	37
21	68	59	49	43
22	75	66	56	49
23	83	73	62	55
24	92	81	69	61
25	101	90	77	68

According to the **table of critical values** for the Wilcoxon's test, for a confidence level of $\alpha = 0.05$ and $N = 14$, the **difference between the two classifiers** is significant if T is ≤ 21 : since $T = 12 < 21$ we can **reject H_0** .

COMPARING MULTIPLE CLASSIFIERS

When dealing with the **comparison of multiple classifiers**, a common example of **questionable procedure** is conducting **all possible paired tests** and reporting results in the form of "model A was found to perform significantly better than B and C, model E performed better than D" and so on.

When so many tests are made, a certain "proportion" of the null hypotheses is rejected due to **random chance**, so **listing them makes little sense**.

Instead, we use the following approach.

Let r_i^j be the *rank* of the j -th of k models on the i -th of N data sets.

The **Friedman test** compares the **average ranks of models**:

$$R_j = \frac{1}{N} \sum_{i=1}^N r_i^j$$

The **null hypothesis** states that **all models are equivalent**.

Thus, if H_0 were true, all the ranks R_j should be equal and the **Friedman statistic** χ_F^2 is distributed according to χ^2 with $k - 1$ *degrees of freedom*:

FRIEDMAN STATISTIC

$$\chi_F^2 = \frac{12N}{k(k+1)} \left[\sum_{j=1}^k R_j^2 - \frac{k(k+1)^2}{4} \right]$$

Notice that $\chi_F^2 \sim \chi^2$ when N and k are *large enough* (as a rule of thumb, $N > 10$ and $k > 5$).

If **number of models and data sets is smaller**, exact critical values have been **computed**.

In any case, Iman and Davenport showed (in 1980) that Friedman's statistic is **undesirably conservative** and derived a better one:

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1) - \chi_F^2}$$

which is distributed according to an **F distribution** with $k - 1$ and $(k - 1)(N - 1)$ **degrees of freedom**. Tables with relevant critical values can be found online or in textbooks.

To better understand this test, let's take a look at an extended version of the example seen in the Wilcoxon's test explanation:

datasets:	C4.5 ↙	C4.5+m	C4.5+cf ↘	C4.5+m+cf
adult (sample)	0.763 (4)	0.768 (3)	0.771 (2)	0.798 (1)
breast cancer	0.599 (1)	0.591 (2)	0.590 (3)	0.569 (4)
breast cancer wisconsin	0.954 (4)	0.971 (1)	0.968 (2)	0.967 (3)
cmc	0.628 (4)	0.661 (1)	0.654 (3)	0.657 (2)
ionosphere	0.882 (4)	0.888 (2)	0.886 (3)	0.898 (1)
iris	0.936 (1)	0.931 (2.5)	0.916 (4)	0.931 (2.5)
liver disorders	0.661 (3)	0.668 (2)	0.609 (4)	0.685 (1)
lung cancer	0.583 (2.5)	0.583 (2.5)	0.563 (4)	0.625 (1)
lymphography	0.775 (4)	0.838 (3)	0.866 (2)	0.875 (1)
mushroom	1.000 (2.5)	1.000 (2.5)	1.000 (2.5)	1.000 (2.5)
primary tumor	0.940 (4)	0.962 (2.5)	0.965 (1)	0.962 (2.5)
rheum	0.619 (3)	0.666 (2)	0.614 (4)	0.669 (1)
voting	0.972 (4)	0.981 (1)	0.975 (2)	0.975 (3)
wine	0.957 (3)	0.978 (1)	0.946 (4)	0.970 (2)
average rank	3.143	2.000	2.893	1.964

We have:

$$\chi_F^2 = \frac{12 \cdot 14}{4 \cdot 5} \left[(3.143^2 + 2.000^2 + 2.893^2 + 1.964^2) - \frac{4 \cdot 5^2}{4} \right] = 9.28$$

And thus:

$$F_F = \frac{13 \cdot 9.28}{14 \cdot 3 - 9.28} = 3.69$$

Looking at an F distribution table with a *significance level* $\alpha = 0.05$ we find that the **critical value** for $F(3, 39)$ is $F^* = 2.85$.

Since we have $F_F > F^*$ we **can reject the null hypothesis**: we have *significant evidence suggesting the classifiers have different performances*.

POST-HOC TESTS

⚠ CLARIFICATION

The test discussed above is used to determine **whether the classifiers perform differently** or not, but it gives **no further information**: it doesn't tell us *which* classifier has the best performance, and neither *how much better* the performance is.

As a result, **after rejecting the null hypothesis**, we need to proceed with **post-hoc tests**.

One such example is given by the **Nemenyi test**.

📖 NEMENYI TEST

Given a *pool* of k classifiers, the **performances of two of them are significantly different** if the *corresponding average ranks differ by at least the critical difference* CD:

$$CD = q_{\alpha} \sqrt{\frac{k(k+1)}{6N}}$$

Where q_{α} are tabulated values (see table below) that depend on the *required significance level* and N is the *number of data sets*.

#	2	3	4	5	6	7	8	9	10
$q_{0.05}$	1.960	2.343	2.569	2.728	2.850	2.949	3.031	3.102	3.164
$q_{0.10}$	1.645	2.052	2.291	2.459	2.589	2.693	2.780	2.855	2.920

Notice that this test works for comparing *two classifiers from a larger pool*: if we're interested in assessing the performances of *all of the classifiers*, as already stated above, reporting the **results of each pairwise comparison is not an acceptable method**.

In this case, it's better to **compare all classifiers** with a **"control" classifier**. Approaches of this kind are aimed at *controlling the family-wise error* in multiple hypothesis testing by making **corrections** that **adjust critical values** to account for the fact that, while using a *control classifier*, we are **making less comparisons** ($k - 1$ instead of $\frac{k(k-1)}{2}$).

🔍 BONFERRONI-DUNN CORRECTION

This correction *controls the family-wise error* by **dividing α by the number of comparisons made** ($k - 1$ in our case).

The corrected critical values are reported in the table below:

#	2	3	4	5	6	7	8	9	10
$q_{0.05}$	1.960	2.241	2.394	2.498	2.576	2.638	2.690	2.724	2.773
$q_{0.10}$	1.645	1.960	2.128	2.241	2.326	2.394	2.450	2.498	2.539

💡 OBSERVATION

The **comparison** between the tables for **Nemenyi's** and **Bonferroni-Dunn's** tests reveal that the **power of the post-hoc test is much greater** when all classifiers are **compared only to a control classifier** instead of between themselves.

In fact, values in the second table are lower: this means that *a smaller value is sufficient to obtain the desired p-value*.

To better understand these tests, let's apply them to the 4 classifiers of the previous example.

Suppose we want $\alpha = 0.05$ significance.

According to the **Nemenyi** test table, we have $q_{0.05} = 2.569$. Thus, the critical difference is:

$$CD_{\text{Nem}} = 2.569 \sqrt{\frac{4 \cdot 5}{6 \cdot 14}} = 1.25$$

In the example, the **best** and **worst performing** classifiers achieved average ranks of 1.964 and 3.143.

Since **even this difference is less than the critical difference**, we conclude that this test is **not powerful enough to detect any significant differences** between the classifiers.

If we instead **choose C4.5** as the **control classifier**, we can use the *Bonferroni-Dunn-corrected values*, in which case the critical difference is:

$$CD_{\text{Bonf-Dunn}} = 2.394 \sqrt{\frac{4 \cdot 5}{6 \cdot 14}} = 1.16$$

With this test, we can conclude that:

- *C4.5+m+cf* performs **significantly better** than *C4.5* ($3.143 - 1.964 = 1.179 > 1.16$).
- *C4.5+cf* does **not perform better** ($3.143 - 2.893 = 0.250 < 1.16$).
- *C4.5+m* is **just below the critical difference** ($3.143 - 2.000 = 1.143 < 1.16$).

BIAS-VARIANCE TRADEOFF

When we talk about model prediction, it's important to understand **prediction errors** due to **bias** and **variance**.

BIAS ERROR

Bias error is an error from **erroneous assumptions** in the learning algorithm. High bias can cause an algorithm to *miss the relevant relations between features and target outputs (underfitting)*. It always leads to **high error on training and test data**.

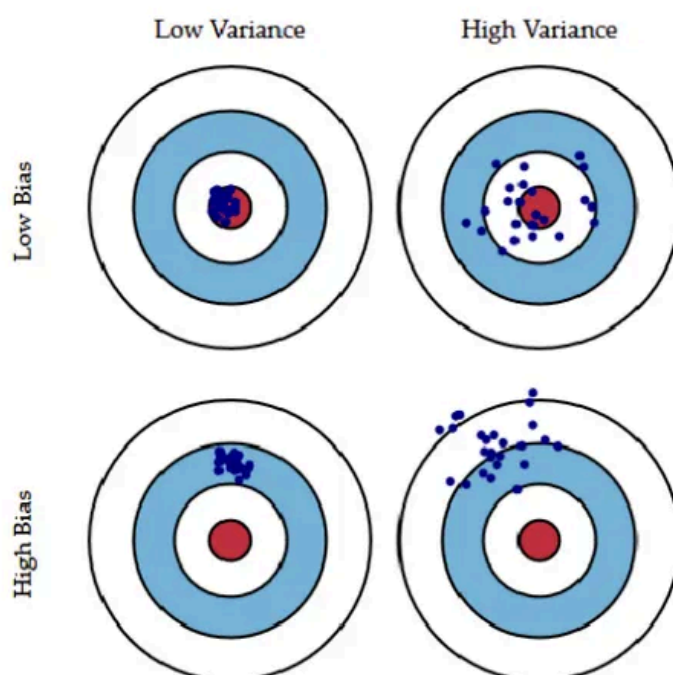
VARIANCE ERROR

The variance is an error from **sensitivity to small fluctuations** in the training set.

High variance can cause an algorithm to **model the random noise in the training data**, rather than the intended outputs (**overfitting**).

As a result, models with high variance **perform very well on training data** but exhibit **higher error rates on test data**.

The figure below helps us understand these errors: *blue dots* represent the model's *predictions*, the *red circle* in the middle represents the *target predictions*.



💡 EXPLANATION

- **High bias** is caused by a **flaw in the model**: since the model itself is **biased**, the predictions will **always show some "minimum" error** with respect to the target.
- **High variance** is a result of **overfitting** the training set: the model **learns the random noise as if it was important** and thus shows **excessive variance** in the prediction outputs.

When we talk about "**bias variance trade-off**" we refer to the problem of **minimising these errors**: it's **not possible to minimise both simultaneously**, hence the term *trade-off*.

BIAS VARIANCE DECOMPOSITION

Before presenting *why we can't minimise both sources of error simultaneously*, we need to understand *how a model's error can be decomposed*.

Let Y be the variable we are *trying to predict* and let X be a *covariate variable*.

The *training set* D consists of a set of points $x_1, \dots, x_n \in X$, each associated with a real

valued *label* $y_1, \dots, y_n \in Y$.

That is, $D = \{(x_1, y_1), \dots, (x_n, y_n)\}$.

We assume there is a **relationship** between the two such that:

$$Y = f(X) + \varepsilon$$

Where ε is **random noise**, distributed with **zero mean** and **variance** σ^2 .

Thus, for each point in the training set we have:

$$y_i = f(x_i) + \varepsilon_i$$

Where ε_i is a *noise sample*.

Our **goal** is to **find a function** $\hat{f}(X)$ that **approximates** f **as well as possible** over all $x \in X$.

To achieve this, we want to **minimise the mean squared error (MSE)** between each y and $\hat{f}(x)$:

$$\text{Err}(x) = \mathbb{E}[(y - \hat{f}(x))^2]$$

In particular, this error can be **decomposed as follows**:

$$\text{Err}(x) = (\text{Bias}[\hat{f}(x)])^2 + \text{Var}[\hat{f}(x)] + \sigma^2$$

Where:

$$\begin{aligned}\text{Bias}[\hat{f}(x)] &= \mathbb{E}[\hat{f}(x) - f(x)] \\ &= \mathbb{E}[\hat{f}(x)] - f(x)\end{aligned}$$

$$\text{Var}_D[\hat{f}(x)] = \mathbb{E}[(\hat{f}(x) - \mathbb{E}[\hat{f}(x)])^2]$$

$$\sigma^2 = \mathbb{E}[(Y - f(x))^2]$$

Notice that *all three terms are non negative*.

🔍 IRREDUCIBLE ERROR

The term σ^2 is **independent of our model**: it appears because of the *random noise on the labels* and is thus defined "**irreducible error**".

✓ DERIVATION

The derivation of the bias variance decomposition for squared error proceeds as follows.

We start by explicitly writing the *mean squared error* of our model:

$$\begin{aligned}
\text{Err}(x) &= \mathbb{E}[(Y - \hat{f}(x))^2] \\
&= \mathbb{E}[(f(x) + \varepsilon - \hat{f}(x))^2] \\
&= \mathbb{E}[(f(x) - \hat{f}(x))^2] + 2\mathbb{E}[(f(x) - \hat{f}(x))\varepsilon] + \mathbb{E}[\varepsilon^2]
\end{aligned}$$

Since ε is **independent** from X , we notice:

$$\mathbb{E}[(f(x) - \hat{f}(x))\varepsilon] = \mathbb{E}[f(x) - \hat{f}(x)]\mathbb{E}[\varepsilon] = 0$$

Furthermore, since ε has **mean zero**, it immediately follows that:

$$\mathbb{E}[\varepsilon^2] = \mathbb{E}[(\varepsilon - 0)^2] = \sigma^2$$

Thus, we need only expand the remaining term:

$$\begin{aligned}
\mathbb{E}[(f(x) - \hat{f}(x))^2] &= \mathbb{E}\left[\left(f(x) - \mathbb{E}[\hat{f}(x)] + \mathbb{E}[\hat{f}(x)] - \hat{f}(x)\right)^2\right] \\
&= \mathbb{E}\left[\left(f(x) - \mathbb{E}[\hat{f}(x)]\right)^2\right] + \mathbb{E}\left[\left(\mathbb{E}[\hat{f}(x)] - \hat{f}(x)\right)^2\right] \\
&\quad + 2\mathbb{E}\left[\left(f(x) - \mathbb{E}[\hat{f}(x)]\right)\left(\mathbb{E}[\hat{f}(x)] - \hat{f}(x)\right)\right]
\end{aligned}$$

We now show that:

$$\begin{aligned}
\mathbb{E}\left[\left(f(x) - \mathbb{E}[\hat{f}(x)]\right)^2\right] &= \mathbb{E}[f(x)^2] - 2\mathbb{E}\left[f(x)\mathbb{E}[\hat{f}(x)]\right] + \mathbb{E}\left[\mathbb{E}[\hat{f}(x)]^2\right] \\
&= f(x)^2 - 2f(x)\mathbb{E}[\hat{f}(x)] + \mathbb{E}[\hat{f}(x)]^2 \\
&= \left(f(x) - \mathbb{E}[\hat{f}(x)]\right)^2
\end{aligned}$$

This comes from the fact that $f(X)$ is **not a random variable**, but a **deterministic function** of the values $x \in X$ (and thus, $\mathbb{E}[f(x)] = f(x)$ and $\mathbb{E}[f(x)^2] = f(x)^2$).

With a similar reasoning, we show that the **last term is null**:

$$\begin{aligned}
&\mathbb{E}\left[\left(f(x) - \mathbb{E}[\hat{f}(x)]\right)\left(\mathbb{E}[\hat{f}(x)] - \hat{f}(x)\right)\right] \\
&= \mathbb{E}\left[f(x)\mathbb{E}[\hat{f}(x)] - f(x)\hat{f}(x) - \mathbb{E}[\hat{f}(x)]^2 + \mathbb{E}[\hat{f}(x)]\hat{f}(x)\right] \\
&= f(x)\mathbb{E}[\hat{f}(x)] - f(x)\mathbb{E}[\hat{f}(x)] - \mathbb{E}[\hat{f}(x)]^2 + \mathbb{E}[\hat{f}(x)]^2 \\
&= 0
\end{aligned}$$

Lastly, we notice that the remaining term can be rewritten as follows:

$$\mathbb{E} \left[\left(\mathbb{E}[\hat{f}(x)] - \hat{f}(x) \right)^2 \right] = \mathbb{E} \left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)] \right)^2 \right]$$

We can finally **plug our derivations back** into the equation and **identify each term**:

$$\text{Err}(\hat{f}) = \underbrace{\left(f(x) - \mathbb{E}[\hat{f}(x)] \right)^2}_{\text{Bias}[\hat{f}(x)]^2} + \underbrace{\mathbb{E} \left[\left(\hat{f}(x) - \mathbb{E}[\hat{f}(x)] \right)^2 \right]}_{\text{Var}[\hat{f}(x)]} + \sigma^2$$

🔍 OBSERVATION

Notice that $\text{Err}(x)$ **depends on the training point** $(x, y) \in D$.

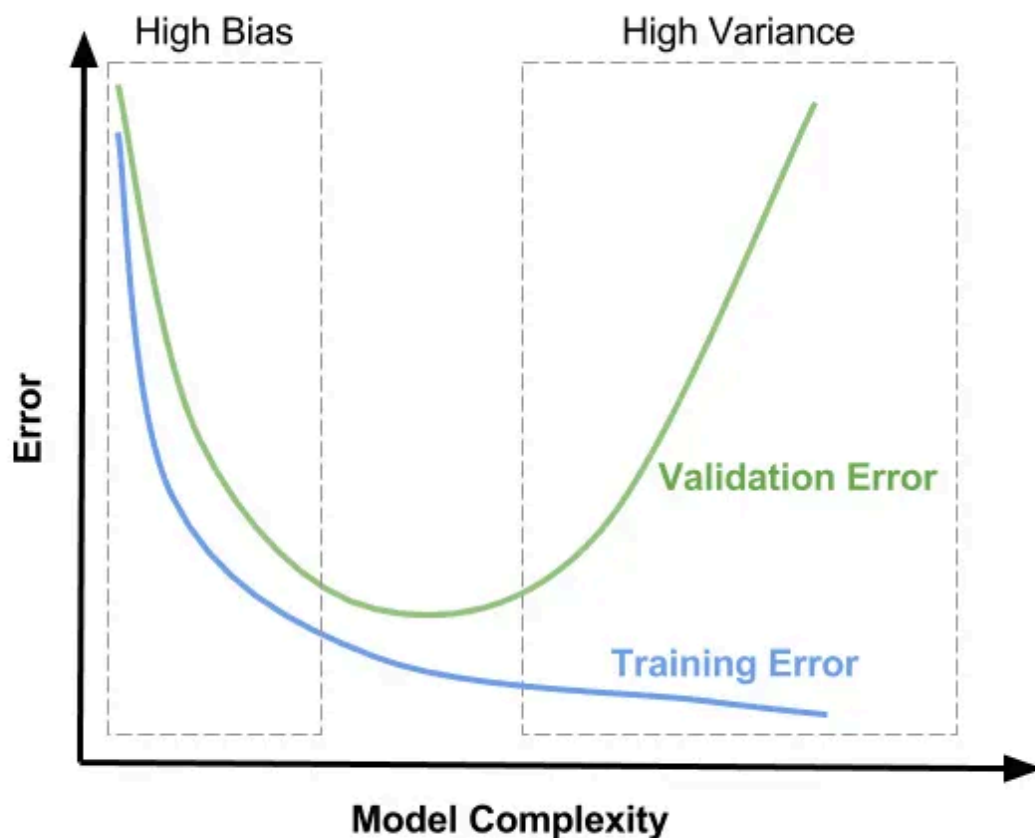
We can compute the **error loss function** by taking the *expectation value over the entire set X* used in the training set D :

$$\text{Err}(\hat{f}) = \mathbb{E}_x \left[\text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)] \right] + \sigma^2$$

OPTIMAL BALANCE

We've seen how the error can be expressed as the **sum** of the model's **bias squared**, the **variance** and an **irreducible error**.

While the **latter** will *always be present*, no matter *how good our model is*, because its **origin lies in the data itself**, the **first two terms can be optimised**.



As seen in the figure above, if our model is **too simple** (e.g. has too few parameters), it may exhibit **high bias** because it *underestimates the complexity of the problem*, and **low variance** because, generally, simpler models produce *more predictable data with less variations around the expected output*.

On the other hand, if our model is **excessively complex** (e.g. has a large number of parameters), it's going to introduce **high variance** and **low bias** because it *perfectly learns the training examples, even the noise in the data*, and thus predictions look **perfectly accurate in the training set**, but **vary unpredictably in the test set**.

The goal is to **optimise this trade off** by finding the *right balance in the model complexity* without overfitting or underfitting the data.